EasyDiffraction and crysta: Diffraction data analysis at ESS



Andrew Sazonov · andrew.sazonov@ess.eu
European Spallation Source (ESS) · Data Management and Scientific Computing (DMSC)

Diffraction

- TOF / CW neutrons and X-rays
- Powder and neutron single crystals
- Commensurate magnetic structures

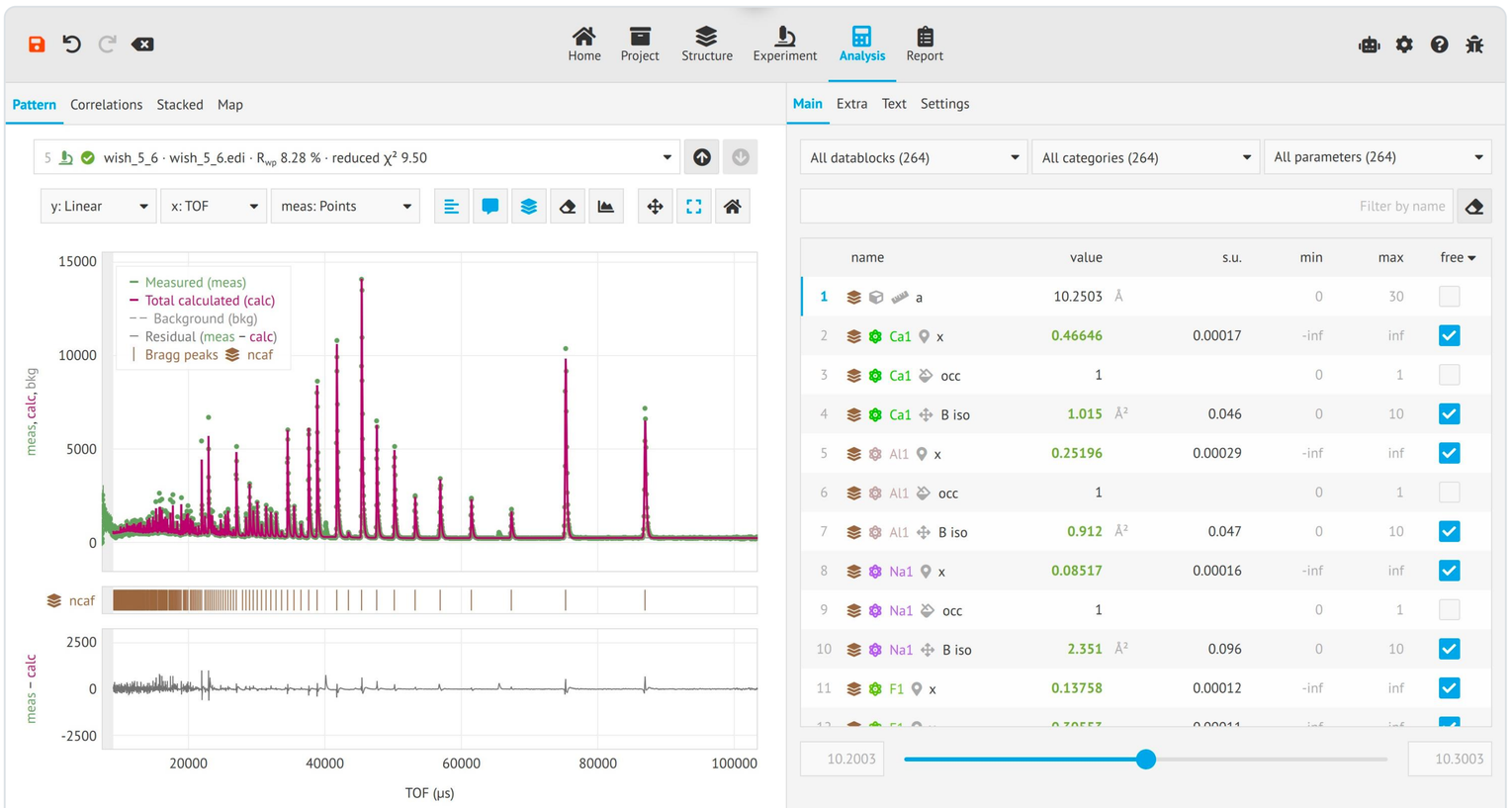
Fitting

- Single and joint refinement
- Sequential and independent scans
- Multiple structures and datasets

One project · three interfaces

- Application · browser and desktop
- Python · scripts and notebooks
- Command line · batch workflows

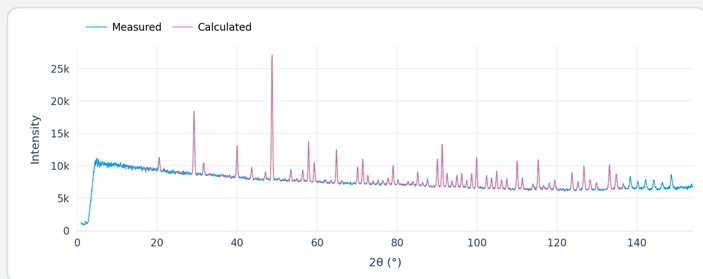
NCAF · WISH · 5 banks · TOF · 193 parameters · Bank 5_6 · Same start



Python / Jupyter · Y_2O_3 · SPODI · AI-ready

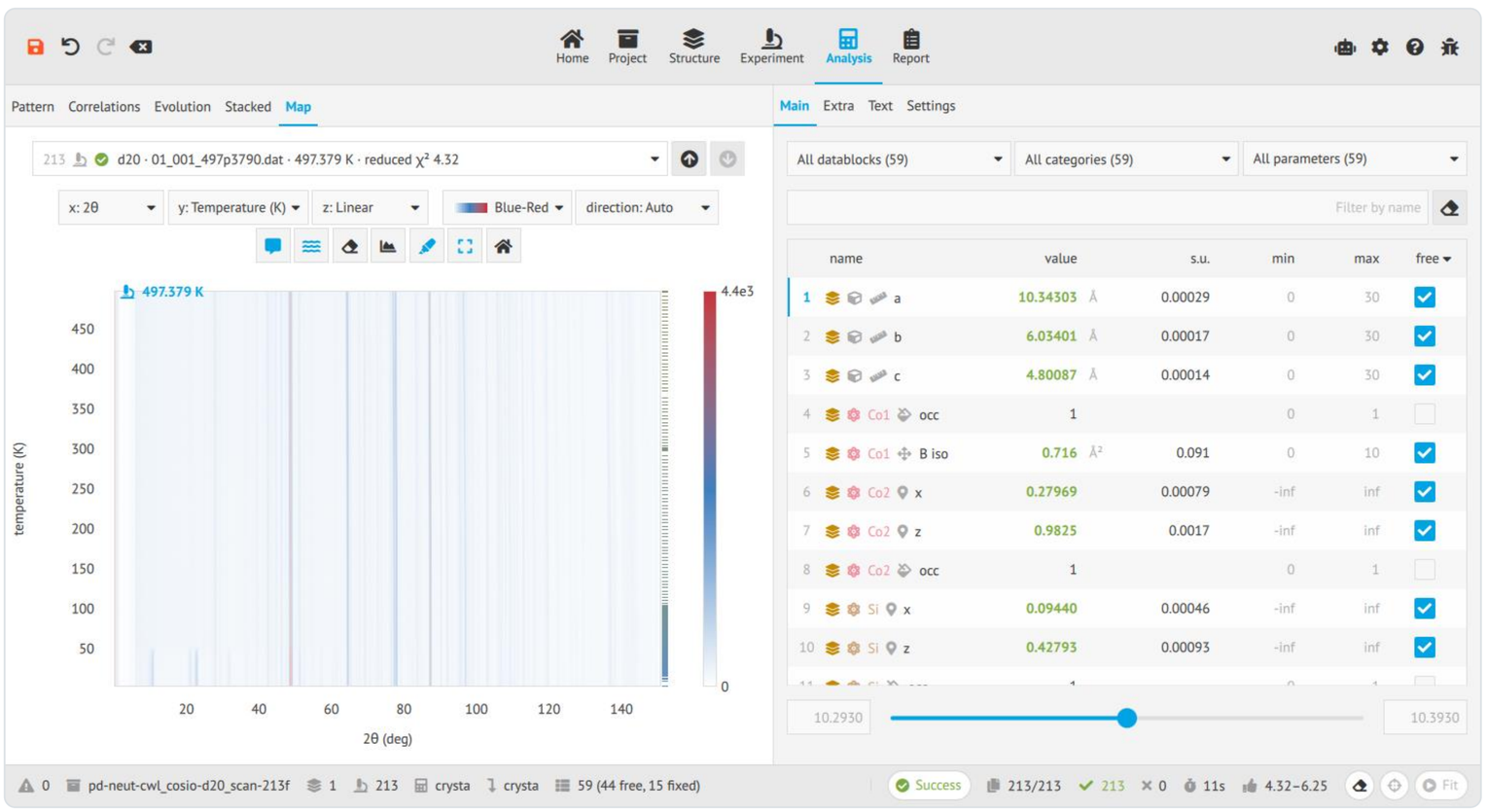
```
In [1]
import edi
case = "pd-neut-cwl_y2o3-spodi"
path = edi.download_data(case)
project = edi.Project.load(path)
```

```
In [2]
result = project.analysis.fit()
print(result.rwp)
```




```
In [3]
project.report.save_html("fit.html")
project.save()
```

Measurement series · Map · Co_2SiO_4 · D20 / ILL · 213 datasets



Speed · FullProf reference

≈3 s crysta · C++20

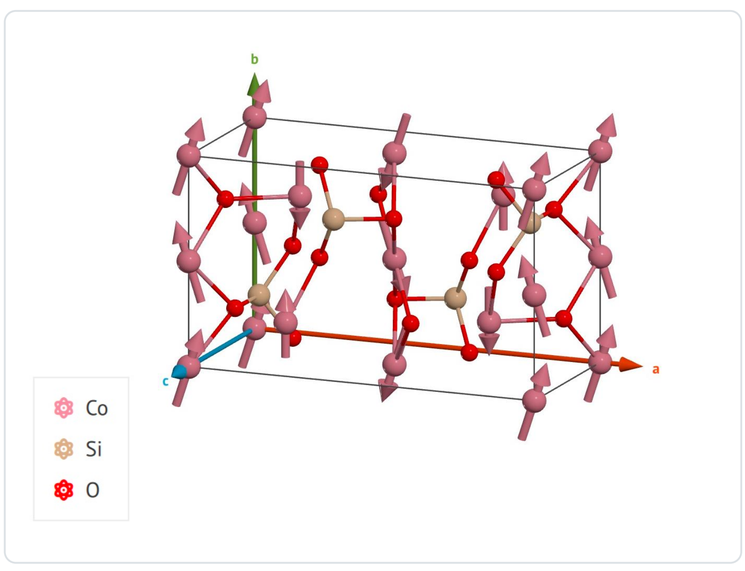
20 s FullProf

NCAF · WISH · 5 banks · 193 p. · Same start · 6-core VISA Linux VM

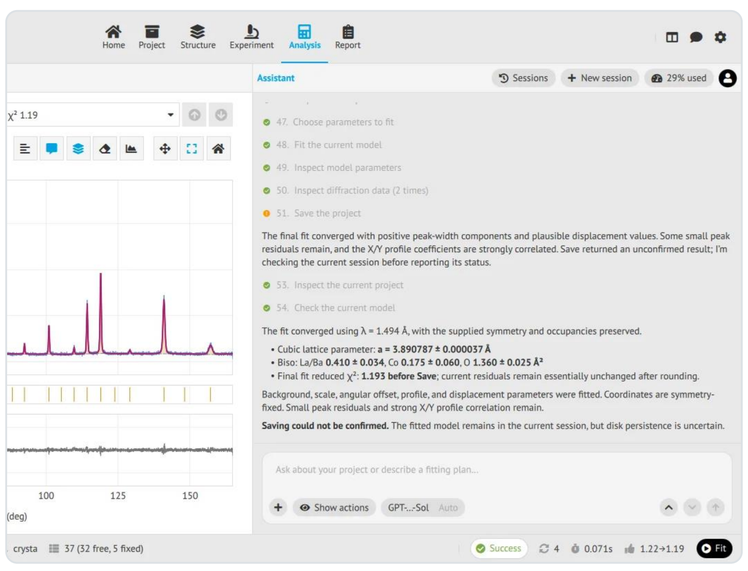
5-bank structural parameters ≤0.3 σ · WISH 2+9 · 48-parameter check · Max Δp 0.69 σ · Median 0.05 σ · Max ΔI≈0.05%

σ: combined standard uncertainty · Unit · integration · system tests

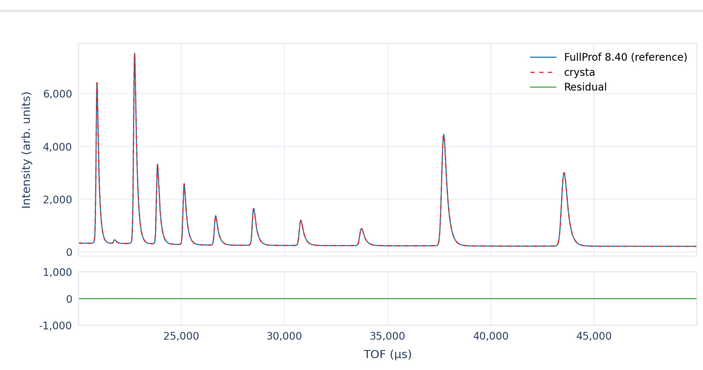
Magnetic fit · Co_2SiO_4 · D20 / ILL · 2.3 K



Desktop AI · LBCO · HRPT / PSI · Fitted



FullProf check · WISH 2+9 · 48 p.



Feedback

https://easydiffraction.org